THE INTERNET & ITS USES

[UNIT 5]

MST_CREATOR

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Note From Author

Hey guys!

I just wanted to tell you that you SHOULD USE OTHER RESOURCES!!!

By that, I am talking about past papers, notes, the book, etc.

You can find these resources from many useful places like r/IGCSE!

You can get many past papers from places like dynamic papers or papa Cambridge.

The Internet & The World Wide Web (WWW)

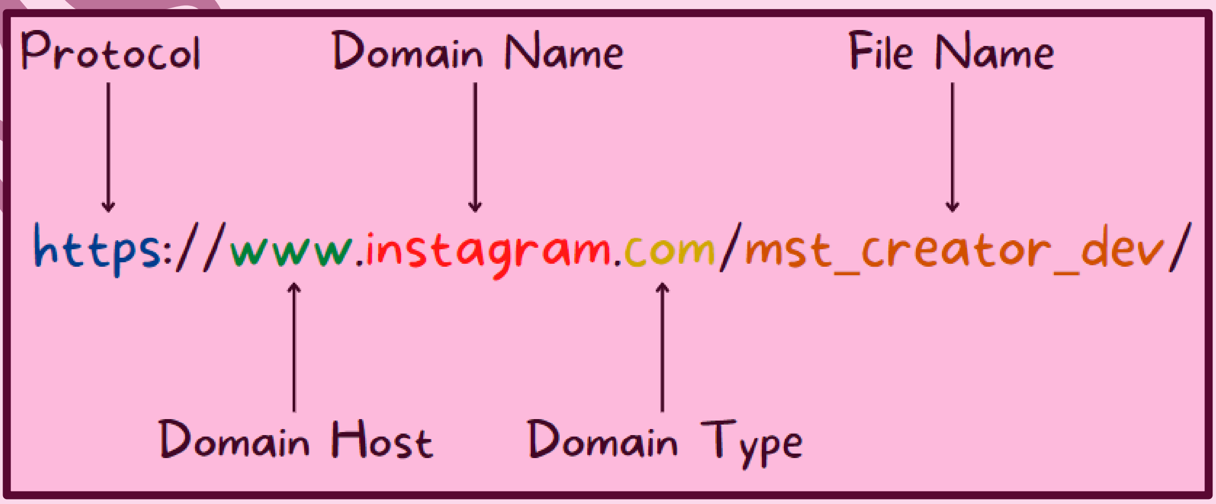
Differences

Internet	World Wide Web (WWW)
System of interconnected computer networks.	Online content (like web pages)
Allow communication between computers (like emails or calls)	HTTP(S) protocols are written using HMTL
Use transmission protocols (TCP) & Internet Protocols (IP)	Use Uniform Resource Locator (URLs) are used to locate web pages.

URLs

A Uniform Resource Locator (URL) is a text-based address which is used to access a website on the internet.

Parts



HTTP & HTTPS

HTTP stands for "Hypertext Transfer Protocol."

HTTP is a set of rules that must be obeyed when transferring files over the internet.

When we use security (like SSL or TLS), we use HTTPS ("S" means Security & indicates a secure way of sending/receiving data over a network).

Web Browsers

A web browser is a software which renders HTML (Hypertext Markup Language) & can display web pages.

Common Features

Feature	Description
Homepage	The page shown when software is opened.
Bookmarks	Stores user's favourite webpages for later use.
User History	Keeps a list of websites visited by the user in chronological order.
Multi-Tabs	Allows user to have multiple tabs open at the same time.
Cookies	Details Here
Navigation Tools	Allows user to go to other opened webpages, pages opened before or after and much more.
Address Bar	This is a box where we can enter the URL of a website and we will be presented with the website.

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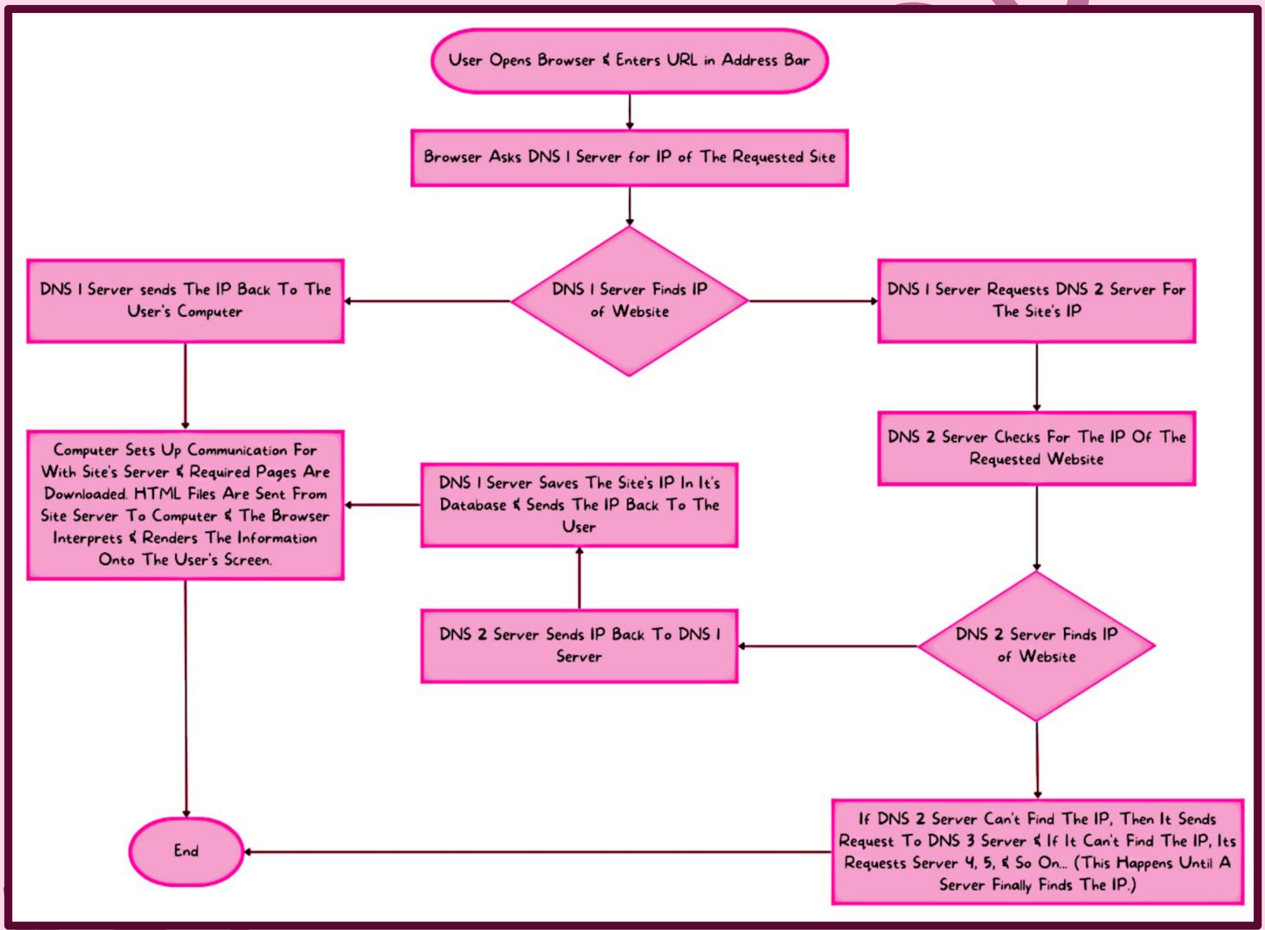
Webpages Location & Retrieval

To get a website, the browser needs to know the IP address, since users cannot remember IP addresses but URLs, we enter that instead, the browser asks a DNS (Domain Name Service) to get the IP.

A DNS is a sort of dictionary with the IP address of many URLs.



Cookies

Cookies are small files or code stored on the user's computer.

Cookies are sent by web servers to a browser on the user's computer.

Cookies store data of a user like previous browser history, currency, language, etc.

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Cookie Types

There are 2 types of cookies:

Session Cookies

These are cookies that stay until the user closes the website or the browser.

These cookies are stored in the temporary memory of the computer.

This type of cookie does not really collect any user information & cannot personally identify the user.

Example: Virtual shopping basket.

Persistent Cookies

These cookies stay on the user's system even after the close the site or browser.

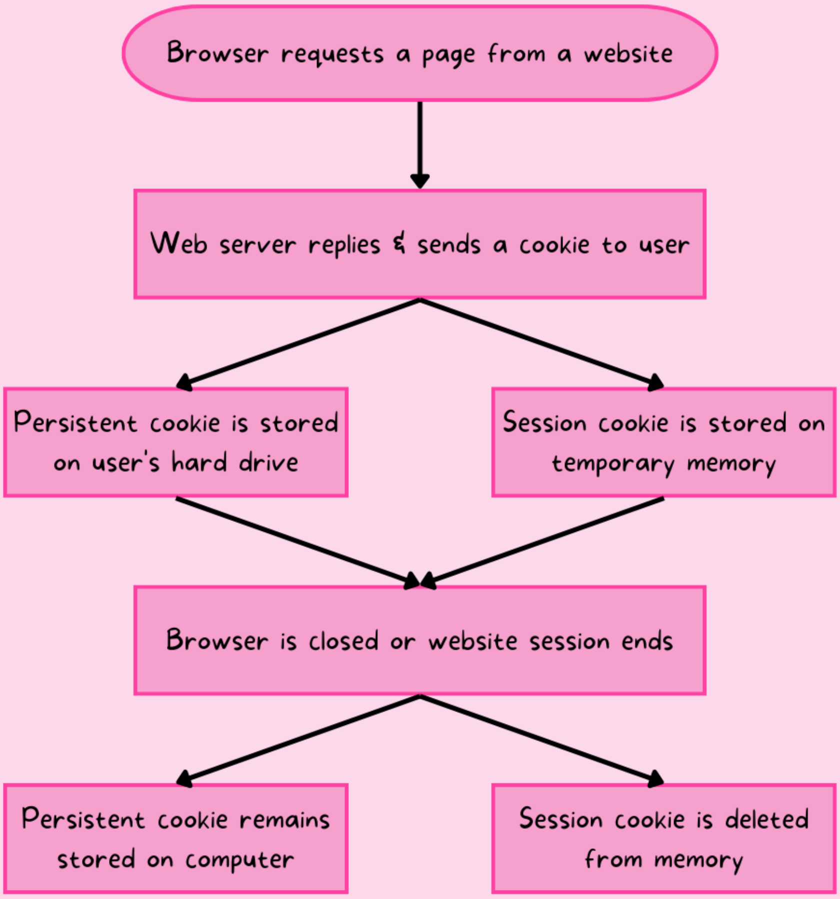
They are stored on the hard drive of the computer & are deleted once they reach their expiry date or the user deletes them.

These cookies save user data like login or personal details.

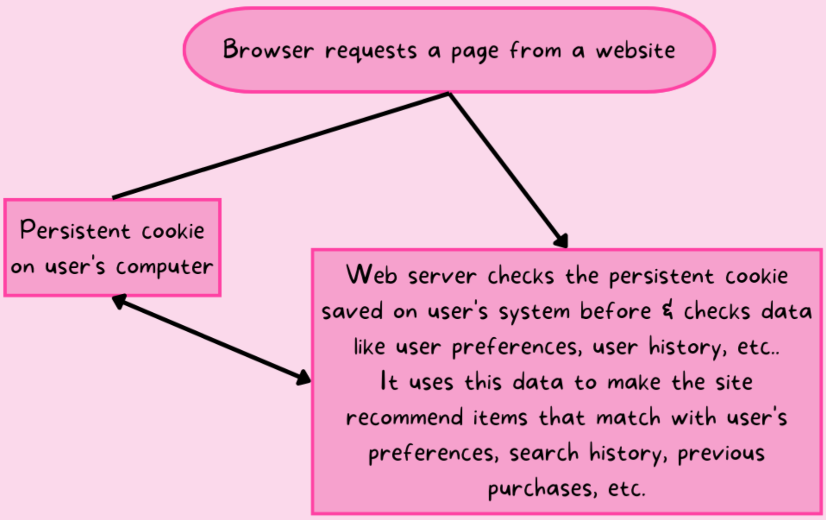
They also store user preferences to recommend them other content which they may like (such as ads or YouTube videos).

Cookies At Work (Diagrams)

When User First Logs In To Website



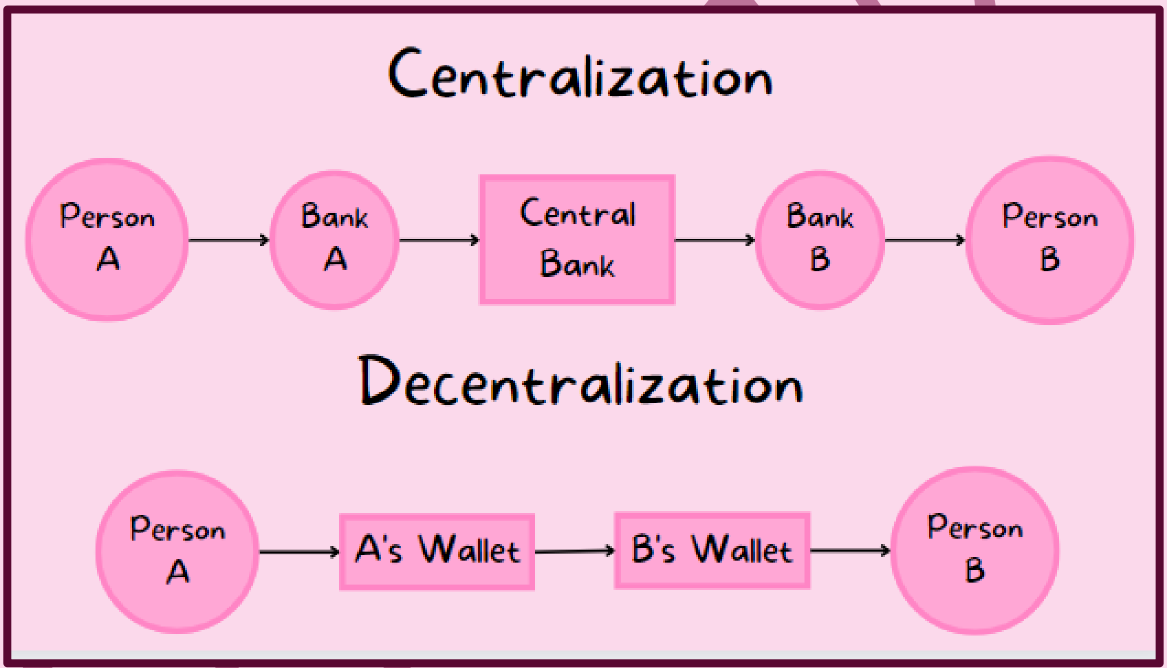
When User Logs Into Website Again



Digital Currency

What Is It?

Digital currency is currency that exists in digital format and is an accepted form of payment to pay for goods & services, it also allows online banking & smartphone payment as it exists as data on a computer system and can be transferred into physical cash when needed.



Centralization	Decentralization
This is when the transactions are processed through the central bank.	This is when the transactions take place directly between the 2 people.

Cryptocurrency uses cryptography to track transactions and was created to address problems with centralization on transactions.

No country controls cryptocurrencies & the rules are set by the cryptocurrency community.

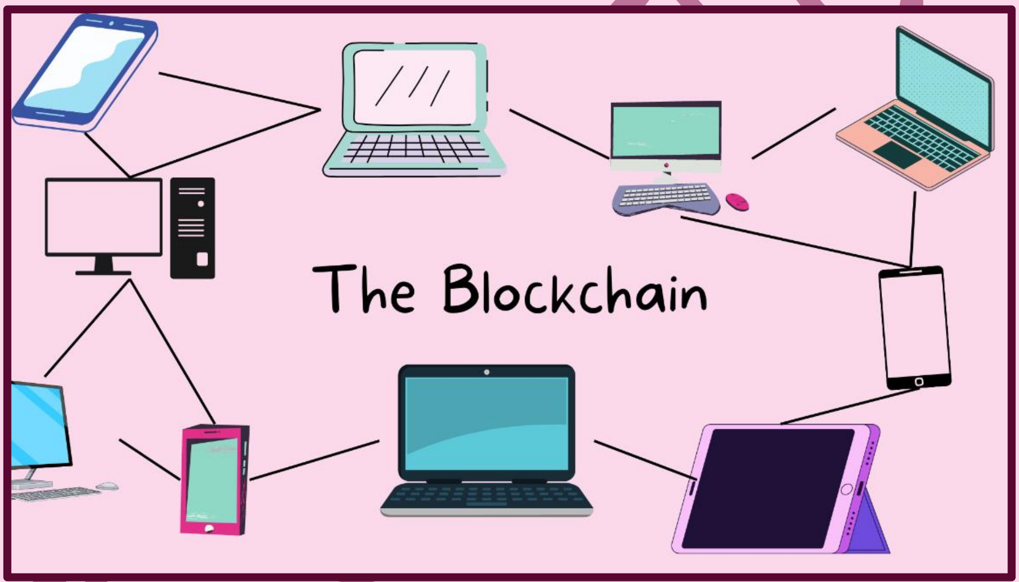
Transactions made using crypto can be tracked and are visible to the public.

The Blockchain

The cryptocurrency system works within the blockchain, this makes it more secure.

The blockchain is a decentralized database.

The blockchain is made up of many interconnected computers which are not connected to a central server, this means that transaction data is stored on all the computers in the network.

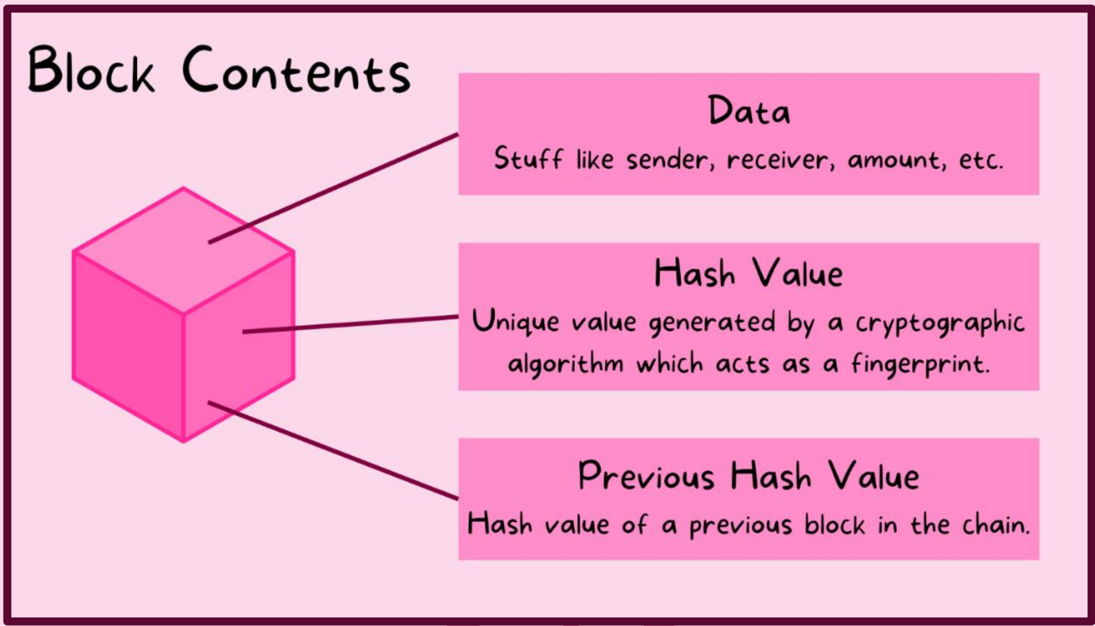


Uses

No.	Use
1	Cryptocurrency exchanges
2	Smart contracts
3	Research (pharmaceutical companies)
4	Politics
5	Education

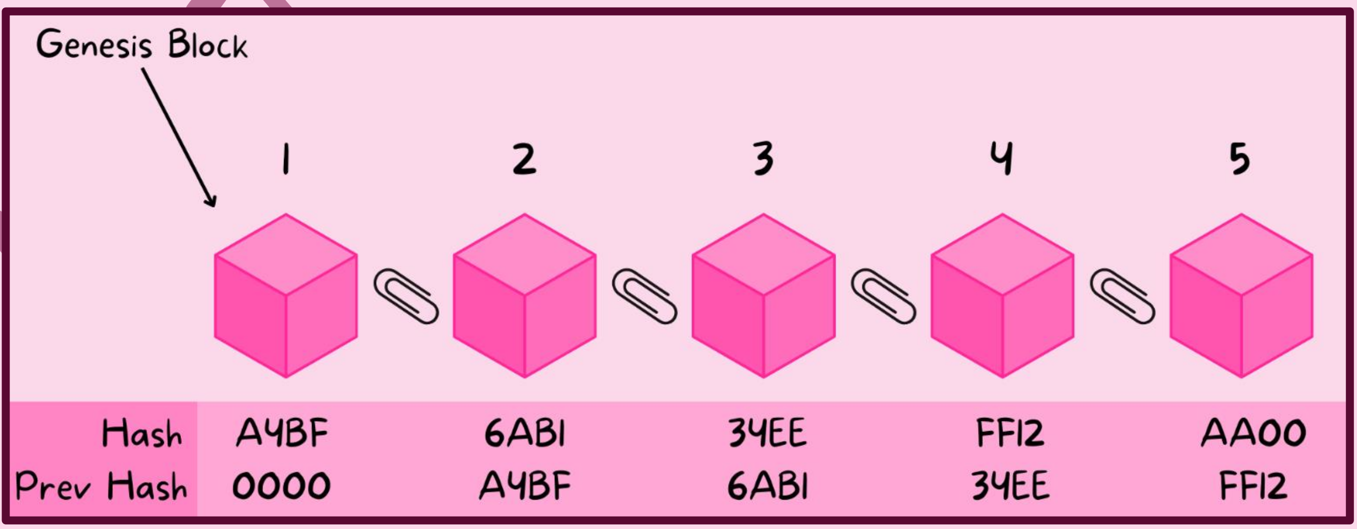
How It Works

Whenever a new transaction takes place, a new block is created.



Every time a block is created, it is given a new hash value and is unique to every block and is given a timestamp (identifies when an event will happen).

Structure

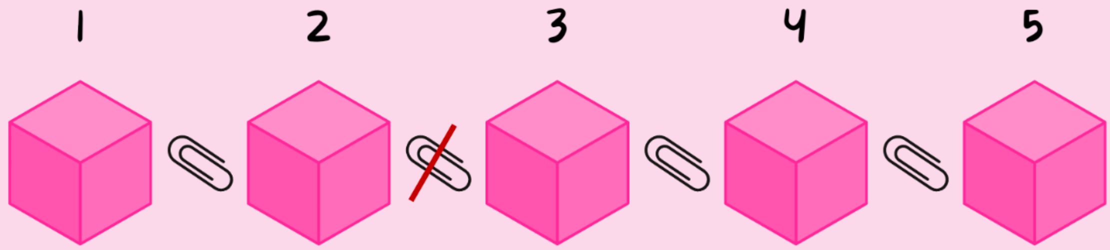


The first block is called the "genesis block."

How This System Is Useful

One thing to know is that the hash of a block changes when the data is changed.

When the hash is changed, the chain of blocks is broken as the previous hash of the next block does not exist.



	1	2	3	4	5
Hash	A4BF	25EE	34EE	FF12	AA00
Prev Hash	0000	A4BF	6AB1	34EE	FF12

Does Not Exist

This prevents people from changing the data.

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Proof of Work

There is still one problem, what if the criminal makes a completely new chain like the tampered one?

Well, proof of work prevents that from happening.

We have some special network users called "miners" who are given a commission for each new block created.

These miners also help regulate the currency.

This means the entire process of chain creation is slowed down which makes it even harder for hackers to tamper any values.

Every miner has a copy of the blockchain system, they get a copy of the newly created block.

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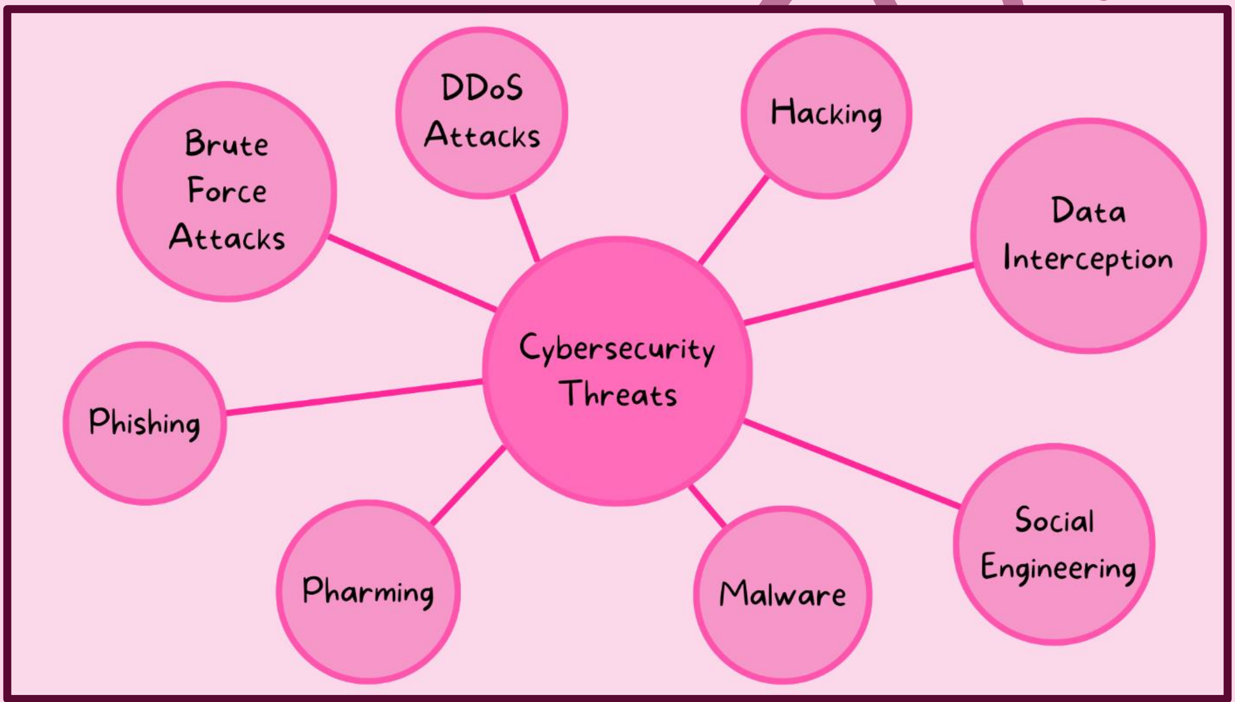
Cyber Security

Cyber Security Threats

We must keep our data safe as it is important to us.

Data can be corrupted or deleted by mistake or through malicious acts.

Cyber security threats are the ways which can cause data to be intercepted



Brute Force Attacks

This is basically when the attacker uses many different combinations of passwords until they eventually get it right.

They usually try using the common passwords to make it faster (like 12345).

They also have a word list which is a text file with many words that may be the password.

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Data Interception

This is when someone tries stealing data by tapping into a wired/wireless connection.

The intent is usually to steal confidential data and compromise security.

How It Works (Wired Connections)

Interception for wired connections is conducted using a packet sniffer which examines the data packets being sent over a network and sends them to the hacker.

*Note: This method is **usually** used for wired connections.*

How It Works (Wireless Connections)

This is conducted using Wardriving or Access Point Mapping.

Data is intercepted with a device (like laptop or smartphone), an antenna, and a GPS (along with some software) outside the victim's house.

The intercepted Wi-Fi signals reveal personal information of the victim without the victim even knowing what is going on.

Prevention

As we know, encryption makes it a bit hard for the hacker to understand the content, we can make it even harder using Wired Equivalency Privacy (WEP) to prevent Wardriving along with a firewall.

We should also use complex passwords for your wireless routers & refrain from using public networks as they have no encryption.

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DDoS & DOS Attacks

DoS Attacks

Denial of Service (DoS) attacks is an attempt of preventing anyone from accessing part of a network (specifically, an internet server).

This is temporary, but it can be a damaging act or a major security breach.

DoS attacks can also happen to users, this prevents them from accessing websites/webpages, online services, and their emails.

DoS Attacks work by sending many requests to a website's server, this causes it to crash as it cannot service the enormous number of requests.

This means that the server will not be able to service the request from a legitimate user.

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DDoS Attacks

A Distributed Denial of Service attack is when many computers send many requests instead of one, this makes it harder to block the attack.

Example

The attacker may send many spam emails to the user.

Internet Service Providers (ISP) provide a certain data quota for all users, so if the attacker sends thousands of emails and fills up the inbox, the data quota will be filled and will not be able to receive any legitimate emails.

Prevention

We can use an up-to-date malware checker.

We can set up a firewall to restrict the traffic to & from a web server or system.

We can also set up email filters to filter out the unwanted emails.

DDoS Signs

There are signs we can check for to see if we have become a victim of DDoS.

Slow network performance is one.

Another is the inability to access certain websites.

The last is that the suer is receiving a huge amount of spam mail.

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Hacking

It is the act of gaining illegal access to a computer without the user's permission.

Hacking can lead to deletion, corruption, or the passing on of data.

It can also cause identity theft.

Encryption makes the data meaningless, but the hacker can still delete it.

Prevention

We can prevent hacking by using firewalls, usernames, strong passwords which are changed regularly, anti-hacking software, and intrusion detection software.

Ethical Hacking

This is when companies authorise paid hackers the check the security measures and how good their computers are when getting hacked.

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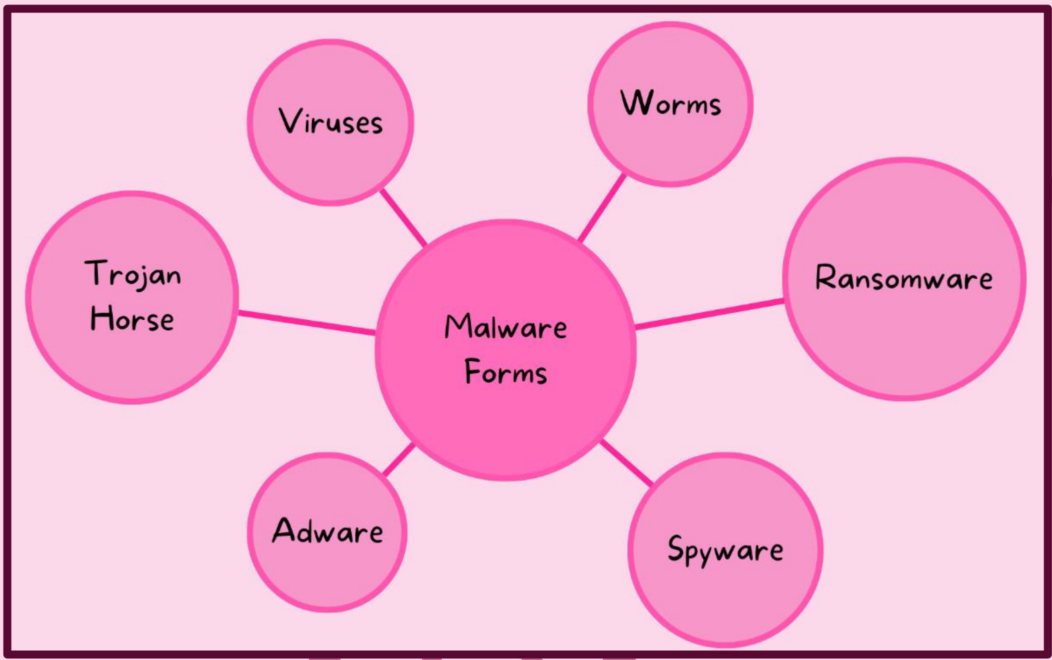
Malware

Malware is software that is specifically designed to disrupt, damage, or gain unauthorized access to a computer system.



Viruses

Identity	
Definition?	They are programs or program code that replicate with the intention of causing a computer malfunction and corrupting or deleting files.
What It Does	They can delete many important files and fill the hard drive with useless data.
Activation	Most viruses need an active host program on the target computer/infected OS. They also have a trigger which causes the virus to start destruction.
Where They Are	Viruses come from many places, it is as email attachments or infected websites or software.
Removal & Prevention	Antivirus software removes them, and firewalls filter them out. Be mindful of what you download and where you download it from.

Worms

Identity	
Definition?	They are a type of standalone malware which can self-replicate and have the intention of spreading and corrupting all the computers on the network.
What It Does	They delete/edit data and inject other malicious software onto a computer. They also take advantage of security failures of the network to travel to other computers on the network.
Activation	No activation is needed, and they don't need an active host program.
Where They Are	Worms can be found as email attachments and message attachments.
Removal & Prevention	Firewalls filter them out & antivirus remove them. Be mindful of what you download.
Examples	ILOVEYOU, Stuxnet, Nimda, Code Red, etc

Trojan Horses

Identity	
Definition?	They are programs which are disguised as legitimate software but have malicious instruction embedded on them and have the intent of doing some harm to the computer.
What It Does	It is designed to damage, disrupt, steal, or inflict dome sort of harm to a user's data or network. They also can give criminals access to the victim's personal information and can function as a gateway to download other malicious programs (like spyware or ransomware).
Activation	Need to be executed by the victim.
Where They Are	Trojans can be found as email attachments and can be downloaded from infected (dodgy) websites.
Removal & Prevention	Antiviruses can remove trojans. Be mindful of what you download & run.
Examples	EGABTR, Back Orifice, ANOM, etc.

Spyware

Identity	
Definition?	This is software that gathers information by monitoring the user's activities on his/her computer.
What It Does	It gathers user interaction on a computer and sends it to the cybercriminal who originally sent the spyware. They are designed to monitor & capture web browsing (and other) activities and capture personal data (like passwords).
Activation	Activates when we run the software it comes along with.
Where They Are	Can come from many places, they can be injected by trojans, downloaded along with software, downloaded from suspicious sites, etc.
Removal & Prevention	Detected & removed by antivirus software. Don't download software from suspicious sites or emails.
Examples	FinSpy, Huntbar, Pegasus, etc.

Adware

Identity	
Definition?	Software which automatically downloads and displays unwanted ads to a user.
What It Does	It can flood a user's screen with unwanted adverts, it can redirect user's browser to a website the adware promotes, etc. It hijacks the browser and makes its own default search requests. It doesn't let antimalware software remove it as it is hard to determine if it is harmful or not. It highlights the weaknesses of a user's defences.
Activation	They are activated when we run the software it comes along with.
Where They Are	Generally downloaded along with software we download from the internet.
Removal & Prevention	Antivirus software can remove it. Be mindful of what you download & where you download it from.
Examples	DeskAd, Look2Me, DollarRevenue, etc.

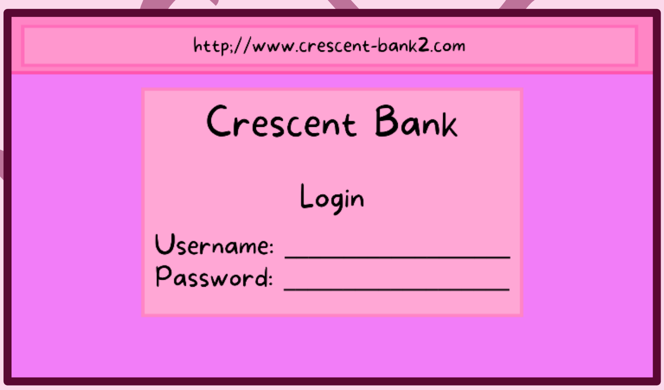
Ransomware

Identity	
Definition?	They are programs that encrypt the data on a user's computer and asks user to pay a ransom to get the decryption key.
What It Does	It encrypts user's files and gives them the decryption key once the cybercriminal's demands have been met.
Activation	No activation is needed.
Where They Are	Comes from Trojans or Social Engineering.
Removal & Prevention	Antimalware/anti-ransomware software quarantines the malicious software. Back up files regularly, antimalware software & monitoring software.
Examples	Bad Rabbit, Reveton, WannaCry, etc.

Phishing

This is when a cybercriminal sends out a legitimate looking email to users.

The email can contain attachment or links which take you to a fake website and can trick you to enter personal information (like credit card number).

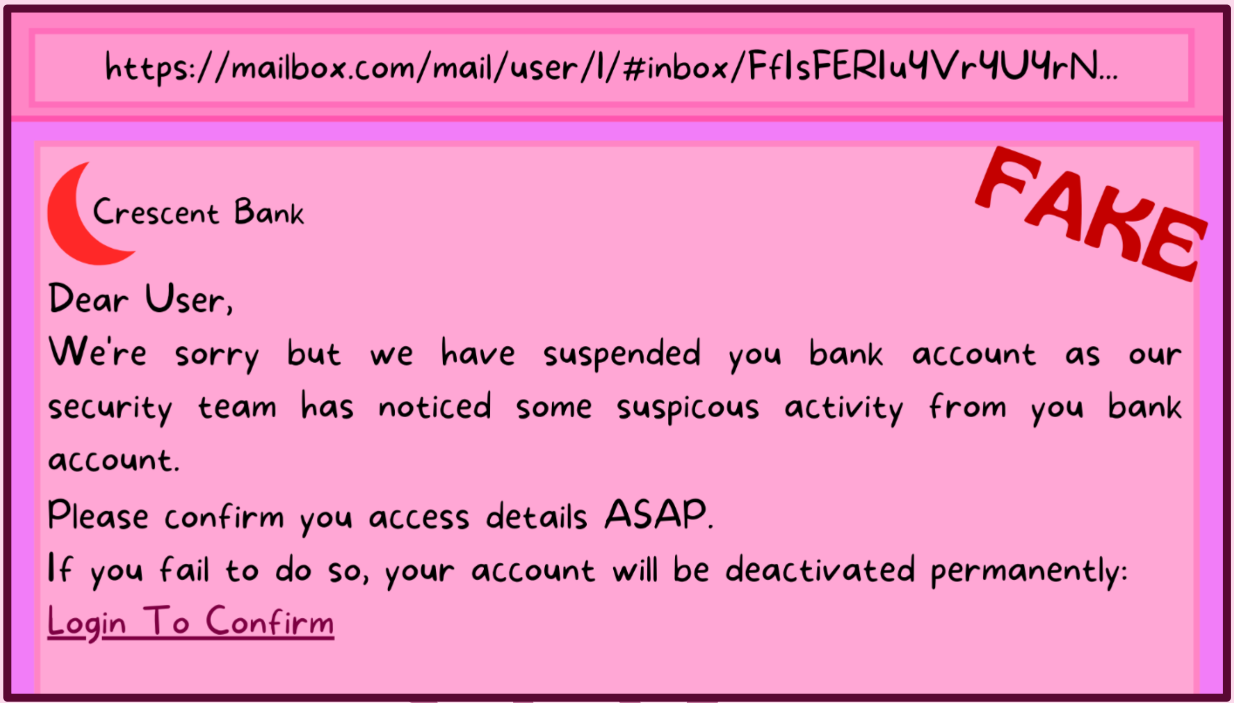


Fake Bank Page

Real Bank Page

The email usually appears to be genuine coming from a known bank or service.

The key point is that the victim must do something so the scam can take place, if the mails are deleted or not opened, they do not inflict any harm.



Example of phishing email.

Prevention

No.	Method
1	Be aware of the new phishing scams!
2	Do not click on any email links unless you know it is safe!
3	Run ant-phishing toolbars on browsers!
4	Check the protocol if it is https or not!
5	Check your online accounts, use strong passwords, and change them regularly!
6	Use up-to-date browser, run good desktop & network firewalls!
7	Make your browser block the popups or close them with the "X" button!

There is also "spear phishing" where cybercriminals target certain users or companies to gain sensitive/confidential information.

Pharming

This is malicious code installed on a user's computer or on an infected website.

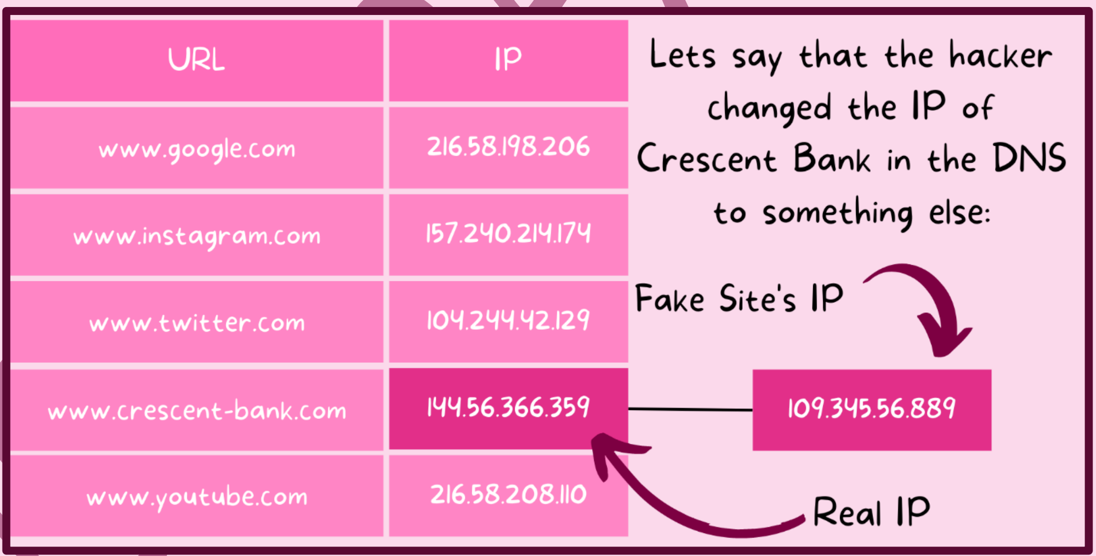
The malicious code redirects the user's browser to a fake website when they enter the URL.

Unlike phishing, pharming does not require user to do anything and can gain all the personal information the user enters onto the website.

How Does It Redirect?

As we know, when the user enters the URL of a legitimate website, the browser redirects the user to a fraudulent one.

This is done through DNS cache poisoning



DNS Server's Database Which Is Edited BY The Hacker

Prevention

No.	Method
1	Use anti-virus software!
2	Check the website spelling & the URL!
3	Ensure that the website uses HTTPs protocol!
4	Read the alerts that your browser shows you!

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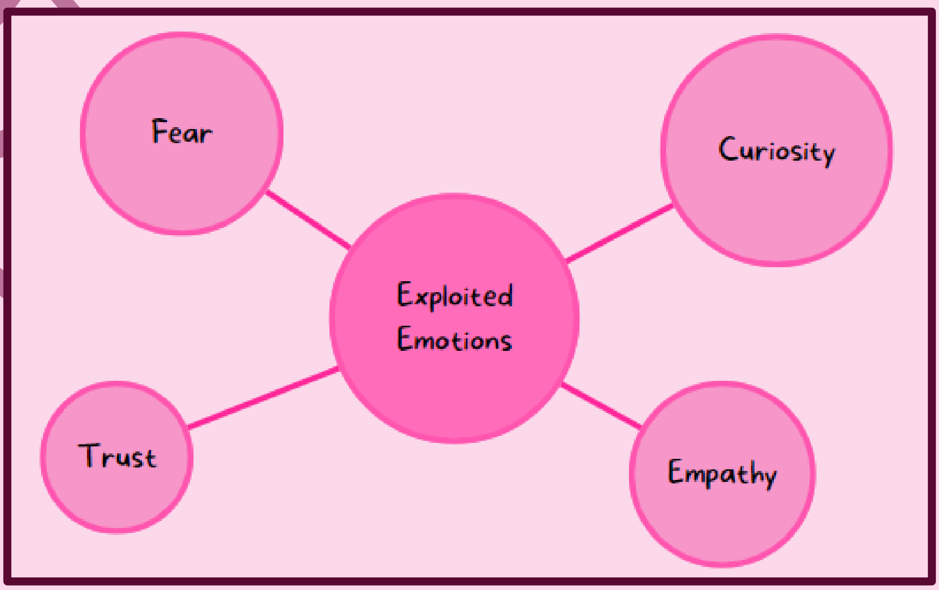
Social Engineering

This is when cybercriminals put the victim into a social situation which causes them to let their guard down and give them their personal information.

Types of Threats

Threat	Description
Instant Messaging	Malicious links are embedded into instant messages that pop up onto the user's screen.
Shareware	Done using popups that persuade the user to accept something which is generally infected.
Emails & Phishing Scams	When user receives a genuine looking email which redirects them to a website which looks genuine but is a fake copy which tricks the user into entering their personal information.
Baiting	This is when the cybercriminal leaves a memory stick (physical) somewhere near the victim who puts it into his/her computer (to see who the owner is) and unknowingly installs malicious software.
Phone Calls	This is when the cybercriminal tricks the victim into giving the criminal control over their device (like those tech support scams).



Emotions cybercriminals exploit

The Process

Victims are identified, information on them is gathered, and attack method is decided.

Victim is targeted (through popups, emails, etc.)

Attack (decided in step 1) takes place and criminal does what they wanted to do.

Criminal tries to remove all traces of malware to cover their tracks.

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Keeping Data Safe

Access Levels

Access levels are basically the different amounts of permission and restrictions users have.

Users with higher access levels have more permission while the ones with less access levels have less permissions.

Uses

Databases

It is important to determine whish users have the permission to read/write/delete the data.

This can be done by having different views of the database (like master view and reader view).

We can also make these views show limited data to users.

Social Networks

Level	Name	Description
1	Public Access	This is the content the public can view.
2	Friends	This is data that can only be viewed by users who are identified as "friends" by the owner.
3	Custom	This allows owner to show data that is viewed by friends but some extra or exclude some data.
4	Data Owner	This is data that can be viewed only by the owner.

The user generally has privacy settings rather than passwords to decide the access levels.

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Anti-Malware

Most common ones are antiviruses and anti-spyware.

You can read about antiviruses on page 4 of my notes of Unit 4 (Software)

Anti-Spyware

This is software that detects & deletes spyware that were illegally installed onto a user's computer.

Methods

[1] Rules

This is when the software looks for certain features which are usually found in spyware & identifies potential risks.

[2] File Structures

This is when the software checks for certain file structures which are commonly found in spyware.

General Features

No.	Feature
1	Detect & remove spyware installed on the device.
2	Prevent user from downloading spyware.
3	Encrypt files to ensure data security.
4	Encrypt key stroke data to remove key logger risks.
5	Block access to user's webcam & mic to not let spyware take over.
6	Scan for signs if user's personal information was stolen & inform user about it.

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Authentication

This refers to the ability of a user to prove who they are.

Usernames & Passwords

Passwords are used to restrict access to data/systems.

Passwords are used on many occasions, like when accessing email account, when banking or shopping online, accessing social media, etc.

Protecting Passwords

No.	Method
1	Use anti-spyware to ensure nobody knows your passwords!
2	Change passwords regularly!
3	Passwords should not be straightforward or easy to crack.
4	Passwords must contain: A number A capital letter A special character At least 8 characters

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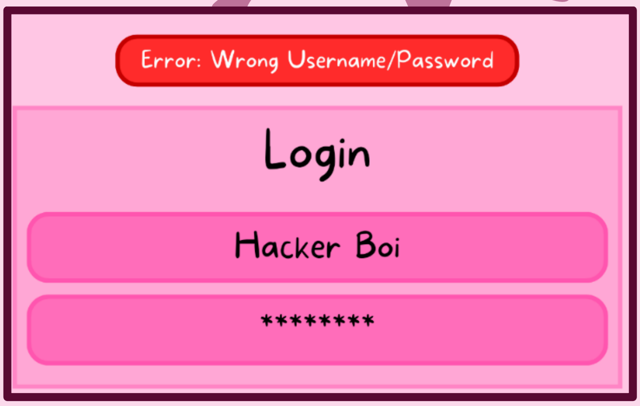
How They Work

All characters in a password are replaced with "*" so nobody can copy the user's password:

Pretty basic login screen

When either the user's password or username is different from the correct data, access is denied & some sort of error appears:



What happens when password or username is incorrect

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When users choose to create a new password, the system tends to ask them to re-enter the new password as verification.

Change Password Prompt

To prevent brute force attacks, users are allowed to enter their password a few times.

Resetting Passwords

If user forgets the password of their account or they want to reset it, they want to reset it.

An email is sent with a link to the webpage to reset their account's password.

This is to prevent unauthorized users from logging into your account.

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Biometrics

Biometrics are like passwords except that they are unique to every human being.

Fingerprints

Fingerprint images are different from person to person.

The image of a fingerprint is compared to the one in the database which was scanned in the past.

The system compares the patterns of ridges & patterns which are unique.

No.	Advantage
1	Improved security as fingerprints are unique.
2	Very useful as fingerprints don't get stolen or lost unlike other devices (i.e., security cards).
3	It is impossible to 'sign in' if there is only one person with the fingerprints of the user.
No.	**Disadvantage**
1	Expensive to install & set up.
2	Scanning accuracy can be affected if the user's finger is injured.
3	People may regard biometric devices as "infringement of civil liberties."

Retina Scans

Retina scans use infrared light to scan the unique blood vessels in the retina.

It is very accurate & is secure.

The only problem is that a person must still for like 10 seconds to scan their retina.

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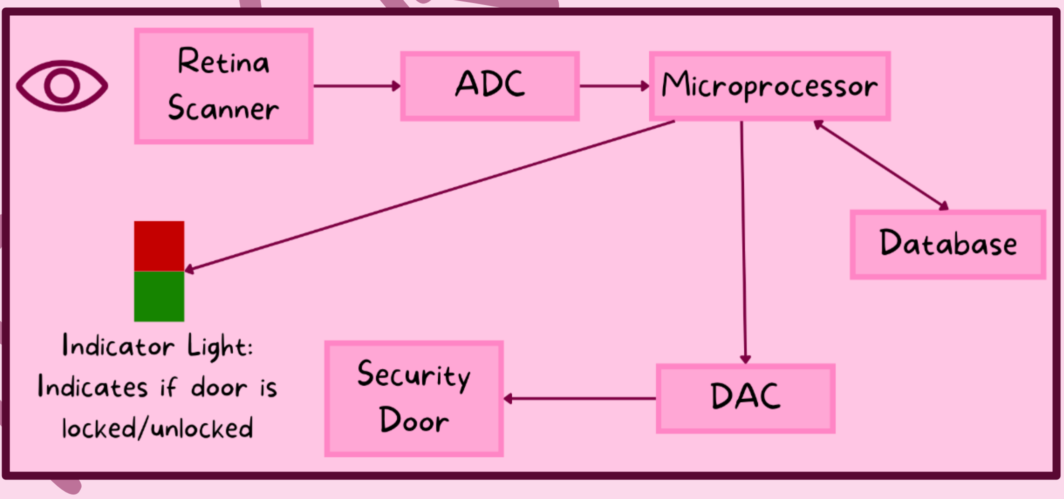
Advantages & Disadvantages of All Biometrics

Technique	Advantages	Disadvantages
Fingerprint Scanning	1. Most developed method. 2. Easy to use 3. Small storage requirement for scanned images	1. Intrusive to some people. 2. Can make mistakes if skin is dry or injured.
Retina Scans	1. High accuracy 2. Hard to replicate someone's retina.	1. Extremely intrusive 2. Slow verification process 3. Expensive to install & setup.
Face Recognition	1. Not intrusive 2. Relatively inexpensive	Result can be affected by lighting, different hair, glasses, age, etc.
Voice Recognition	1. Not intrusive 2. Fast verification method 3. Relatively inexpensive	1. Someone can record owner's voice and gain access 2. Low accuracy 3. Hard to identify voice of owner when ill.

Example With Retina Scanner



A person stands facing the retina scanner which sends the scanned data to the ADC which coverts this data to digital data and then goes t a microprocessor which compares the scanned data with the data stored in the database. If the 2 data sets match, the microprocessor sends signal to turn the green light on and unlock the door, the door is controlled by a DAC & an actuator. On the other hand, if the data sets do not match, then access is denied, and the light remains red. *(Memorize this)*

2-Step Verification

This requires 2 methods of authentication to verify a user.

Mainly used when user makes online purchases using credit/debit card payment.

Example [Anime Character btw]

Let's say Alucard decides to buy some little addons for his Jackal from a website.

Alucard would log in to the website using his username and password (step 1).

To improve security, an 8-digit pin (one time code) is sent to him via text message or email.

Alucard enters the pin into the website and is now authorised to purchase parts for Jackal.

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Automatic Software Updates

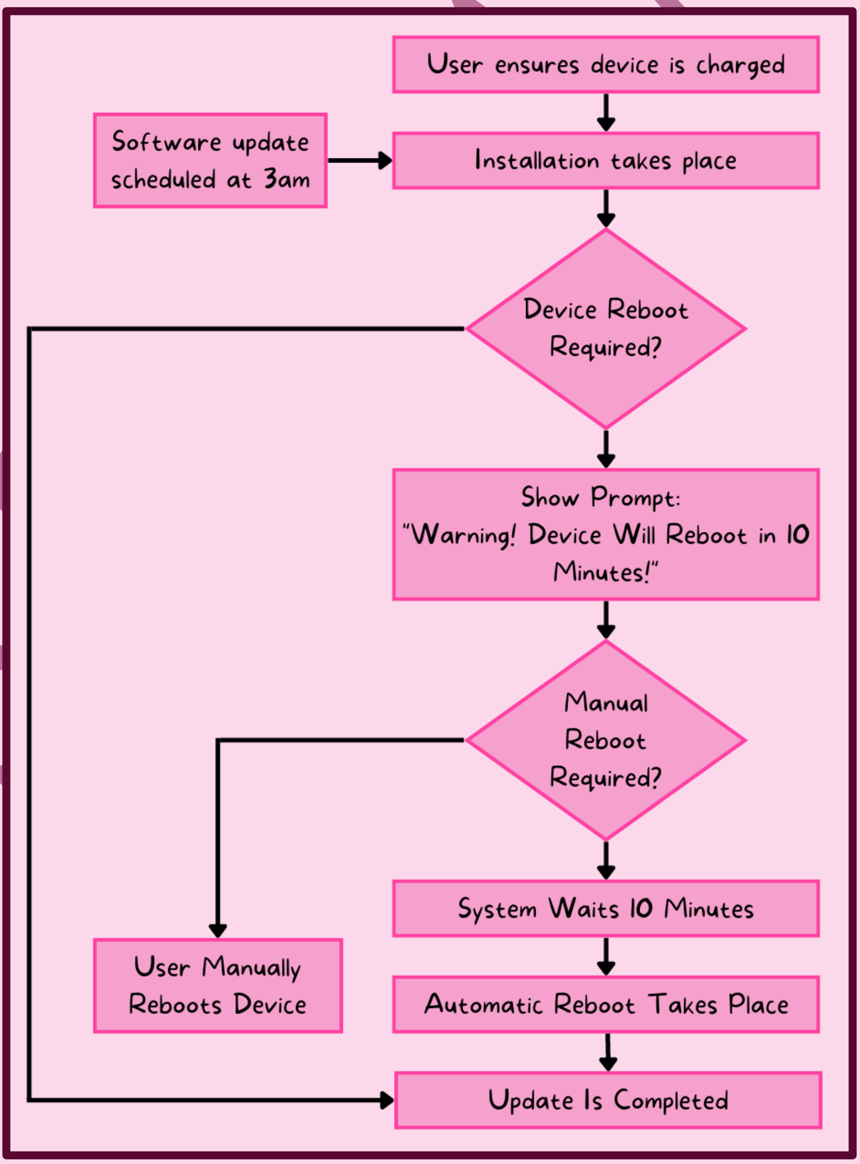
This is when software on devices is kept up to date.

This is done overnight or when you are logged off the device.

These updates are essential as they contain patches to improve the software's features, fix bugs, prevent irregular occurrences, and improve security.

The only disadvantage is that it may disrupt your device after installation.

This can be fixed by downloading another patch or just restore the older version of the software.



Check Spelling & Tone of Communication & URL Links

Before You Open Any Links

1. Check for spellings & grammar in email and URL of links.
2. Check the tone of the email, rushed emails tend to be scams
3. Check the email address itself and check if it is legitimate

The Links

1. Check if the website is using HTTPs protocol
2. Check for typos in the URL as the name may have unnoticable differences (Typo Squatting)
3. Check if the link has anything to do with the company sending the mail.

Firewalls

A firewall is either software or hardware

It sits between user's system and external network (like the internet).

It filters information in and out of your computer.

Firewalls defend against phishing, pharming, malware, hackers, etc.

They let user decide if they want communication with external sources and notifies user when external source is attempting to access his/her system.

Tasks

No.	Task
1	Examine traffic between an internal network & an external network
2	Checks if incoming & outgoing data meets a set criteria.
3	Traffic that fails to meet the criteria is blocked & user is warned about security issues
4	Firewall is used to log all incoming & outgoing traffic for later interrogation
5	Criteria can be set to prevent user from visiting undesirable sites and has a list of the blocked Ips
6	Firewalls help prevent malware and hackers from entering the computer
7	Warns user if a software they have is trying to access external data sources and is given choice to allow this or not

Times When Firewall Cannot Block Harmful Traffic

No.	Description
1	Cannot prevent users on the internal network from bypassing firewall
2	Employee carelessness or misconduct cannot be prevented by firewall
3	Firewalls cannot do anything if user disables them

Proxy Servers

No.	Feature
1	Allows traffic to be filtered and can block sites if needed
2	Keep user's IP secret to improve security
3	Access to web server is allowed if the traffic is valid
4	Access to web server is denied if the traffic is invalid
5	Possible to block requests from certain IP addresses
6	Prevents direct web server access
7	Launched attacks hit the proxy server rather than user's computer
8	Invalid traffic is directed away from web servers which adds protection
9	Cache allows webpages to load faster (they are downloaded on user's first visit)
10	Function as firewalls

Privacy Settings

They are controls available on web browsers, social networks, etc.

They are designed to limit who can access user's personal profile.

General Features

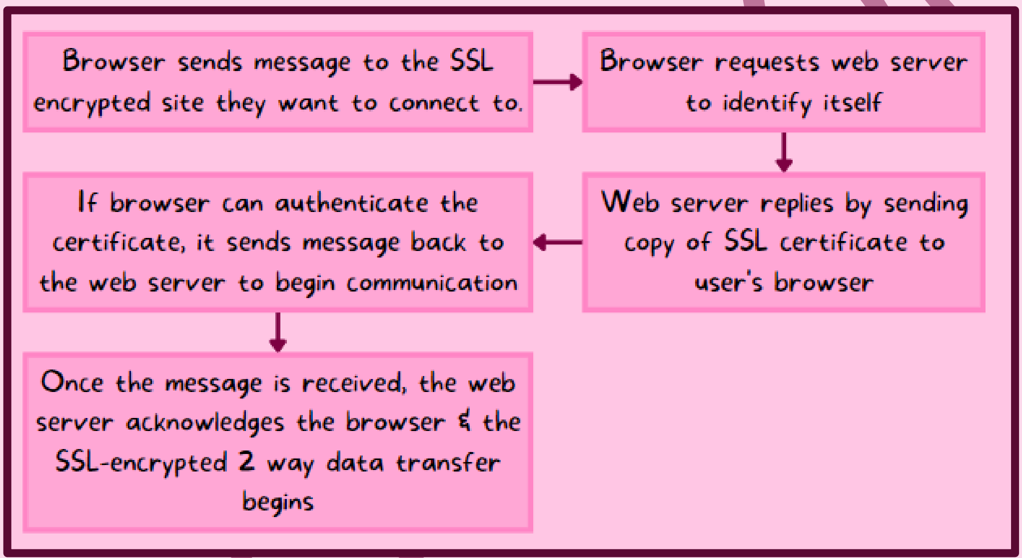
No.	Feature
1	'Do Not Track' option prevents sites from collecting and using browsing data
2	Check to see if payment methods were saved on site, this prevents user from typing payment details every time
3	Safer Browsing, this means that browser alerts user if they opened a dangerous website
4	Web browser privacy options (like history and cookies)
5	Web advertising opt-outs, this lets us prevent third parties from accessing or browsing data for advertising purposes
6	Apps, to let or not let sites access information (like location)

Secure Sockets Layer (SSL)

This is a type of protocol (set of rules used by computers for communication) which allows data to be transferred securely over the internet.

Websites encrypt data with SSL, this means that only the user and the web server understand the data.

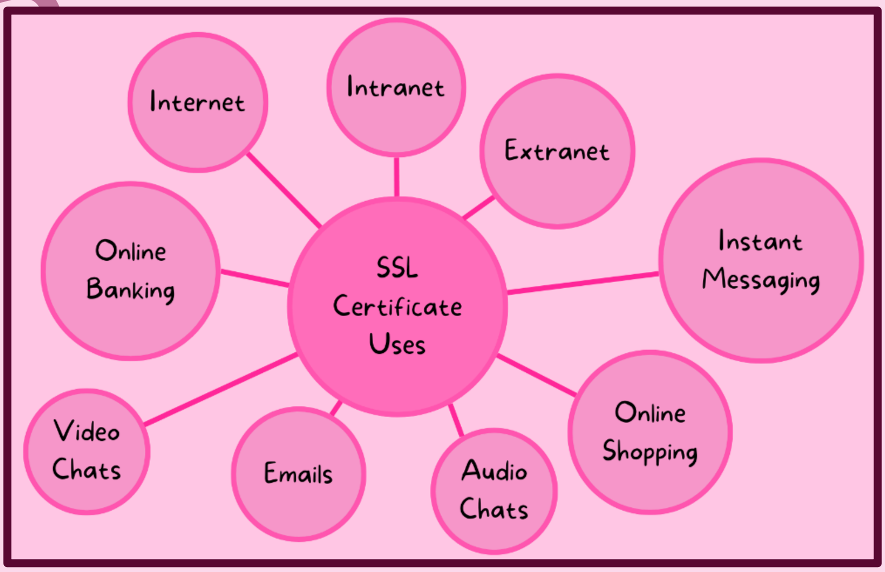
When SSL is used, the padlock is present.



SSL Certificates

These are digital certificates which are used to authenticate websites.

If the certificate is authenticated, then any communication/data transfer is secure.



Thanks For Reading!

If These Notes Helped in Any Way…

I'm Happy for You 😊

Note: All Diagrams Are by Me

MST_CREATOR